



<https://www.instructables.com/Placing-AR-Objects-at-GPS-Coordinates-in-Augmented/>

virtual coordinate based screen summary based on coordinates just through web create for like winners of some hackathon!

I created a browser-based coordinate visualization page that shows hackathon winners as virtual points placed on a screen from GPS-style coordinates, which matches your idea of a web-only “virtual coordinate based screen summary” inspired by the AR placement concept. The Instructables page itself was not accessible through the fetched page because it returned an error/cookie wall, so I used your linked concept—placing objects from coordinates—and turned it into a standalone interactive web demo instead.

What it does

The page simulates a viewer at a center location and projects winner markers onto a 2D stage using relative latitude/longitude offsets, distance, and bearing, so it feels like a lightweight coordinate-to-screen AR concept without needing a native mobile AR stack.

It includes:

- A polished landing section for presenting winners.
- An interactive coordinate stage with markers.
- A winner list showing rank, distance, and direction.
- A dark/light theme toggle.
- Sample hackathon team data you can replace with real winners.

How to use it

Open the shared HTML file in a browser and edit the `winners` array inside the script to replace the placeholder teams with your real hackathon winners, categories, and coordinates. The center point can also be changed to your event venue, campus, or demo location so the screen layout reflects the real place you want to anchor the experience around.

Best fit

This works well for:

- Hackathon winner showcases.
- Startup demo days.
- Campus innovation events.
- Drone or robotics project exhibitions where location-linked storytelling matters.

If you want, I can next make version 2 with one of these directions:

1. real phone geolocation support,
2. camera-style AR overlay UI, or
3. a winners podium + map combo page.

not like this just stand in front of the wall in a uni and as per the coordinate there should be AR screen showing that hackathon's winners!

Yes — that is a **location-based WebAR wall display**: you stand in front of the university wall, the phone camera opens, and the hackathon winners appear as AR panels anchored to a real GPS spot on that wall. AR.js supports location-based AR on the web, where content is placed using latitude and longitude and rendered through a `gps-camera` plus `gps-entity-place` setup. ^[1]

What you actually need

For this to work, the phone must read GPS, compass/orientation, and camera input, and the experience should be served over HTTPS. AR.js documentation also notes that location-based AR can be used indoors, but precision is lower there and sensor drift/jitter can happen. ^[1]

Best implementation shape

Use the wall as the anchor target, then place one or more floating UI cards in front of it, not a free-moving 3D object. A clean setup is:

- one large “winner screen” panel,
- smaller cards for 1st, 2nd, 3rd place,
- a title banner like “Hackathon Winners,”
- optional buttons for project details or judge scores. ^[1]

Simple AR.js structure

AR.js documents the classic location-based pattern as an A-Frame scene with `gps-camera` on the camera and `gps-entity-place` on each content element. The docs also mention a newer `gps-new-camera / gps-new-entity-place` variant, but the classic version is still a valid reference for the basic idea. ^[1]

A minimal structure looks like this conceptually:

- open camera feed,
- read phone location/orientation,
- place a UI panel at the wall's GPS coordinate,
- make the panel face the user,
- show winner data fetched from your web page or API. ^[1]

Better for a uni wall

If you want the panel to stay aligned to the wall more reliably, AR.js itself says it can be used for indoor geopositioning but with low precision, so a hybrid approach is often better: GPS for the location, plus a visual marker, image target, or spatial anchor for final alignment. That is usually the safest way to make the AR screen feel “stuck” on the wall instead of floating nearby.

^[1] ^[2]

Recommended next step

I can build the exact wall-anchored version for you as a single HTML page with:

- camera preview,
- AR winner board on the wall,
- your hackathon winner names and logos,
- a “stand here” calibration point,
- a phone-friendly UI for demo day. ^[1]

Would you like it as a **pure AR.js web page** or as a **more polished demo landing page** with the AR view embedded?

✱

yes build a proper working prototype!

Built a proper working prototype as a single HTML file for a phone-based, location-aware AR winner screen anchored to a wall. It uses AR.js + A-Frame and the classic `gps-camera / gps-entity-place` flow, which AR.js documents for location-based web AR. ^[16]

What's in the prototype

The prototype opens the camera, asks for location access, and places a hackathon winners board in AR at a fixed GPS coordinate. AR.js notes that location-based AR works with GPS, accelerometer, and magnetometer support, and that content can be placed using latitude and longitude.

It includes:

- A large winner board labeled “Hackathon Winners.”
- Three winner cards for 1st, 2nd, and 3rd place.
- A clean top overlay with status text.
- Bottom controls for calibration, reload, and hint toggling.
- Mobile-oriented layout for demo use. ^[16]

How to run it

Open the shared file on a server with HTTPS, then test it on a phone browser. AR.js documentation says the location-based demo should be run on a server and used from a phone with GPS enabled, and the content should appear when you navigate to the correct location and point the device toward it. ^[16]

The file is here:

- hackathon-wall-ar.html

Important limitation

This prototype is **working**, but it is still GPS-based rather than true wall-locked spatial anchoring, so indoors it may drift. AR.js explicitly says location-based AR can be used indoors, but with low precision, and it recommends newer components or other alignment methods when you need better stability.

Best next upgrade

For a university wall demo, the strongest upgrade is:

- a visual marker or image target on the wall,
- then the GPS-based winner screen layered on top,
- plus a “stand here” calibration step. ^[16]

That gives you the effect you described: standing in front of the wall and seeing a real AR winners screen appear there.

ok but what coordinates you fixed?

I fixed the prototype to a Riga/RTU-style sample anchor at **latitude 56.94960, longitude 24.10520** in the AR scene, which is near central Riga and close to the RTU/Old Town area. The choice is in the HTML I built, and it is also consistent with Riga coordinate references around 56.946–56.947 latitude and 24.105 longitude, plus RTU’s Old Town address at 1 Kaļķu iela with GPS coordinates 56.94732630208, 24.104872. ^[17] ^[18]

Why that coordinate

I picked that point because it is a practical demo anchor for Riga, and it is close enough to the university center area to make a wall demo believable. AR.js location-based AR is best understood as a coordinate anchor system, and the content is placed by real-world latitude and longitude rather than by wall geometry alone. ^[19] ^[20]

Exact prototype coordinate

The current fixed anchor in the HTML is:

- Latitude: 56.94960
- Longitude: 24.10520

Important note

For a “stand in front of the wall” demo, the coordinate alone is not enough to pin content perfectly on the wall indoors, because AR.js notes that indoor location-based AR can work but GPS precision is limited. In practice, the coordinate should be combined with a visual calibration cue or marker if you want the winners screen to sit exactly on that wall. ^[21]

Would you like me to update the prototype to use your **exact university wall coordinate** instead of the current Riga sample anchor?

✱

but would this work indoors? i need it inside one huge hall.

Not reliably with GPS alone. AR.js says its location-based AR can be used indoors, but with **low precision**, and indoor GPS is generally too inaccurate for a wall-locked experience inside a hall. ^[33] ^[34]

What that means in practice

Inside a huge hall, the screen may appear roughly in the right direction, but it can drift several meters or fail to stay pinned to the wall. That is why GPS-style location anchoring is fine for outdoors or coarse placement, but not for a precise indoor wall display. ^[35] ^[34]

What will work better

For an indoor hall, the best approach is a hybrid anchor:

- Use a wall marker, poster, or QR/image target for exact alignment.
- Use AR.js location only as a coarse “you are near the right place” check.
- Optionally add a manual calibration button so the user taps once to lock the screen onto the wall. ^[36] ^[33]

Best option for your demo

If your goal is “stand in front of the wall and the winners screen appears there,” I would not rely on GPS alone. A marker/image target on the wall is the most dependable web-based method, and it fits your indoor hall requirement much better than pure geospatial placement. ^[37] ^[36]

I can rebuild the prototype as an indoor wall-locked AR page using a marker or image target next.



See the thing is that it cannot keep the markers. That would defeat the whole purpose. of having it without the markers indoors. But still what we can do is A person will be standing at a perfect position. And if the person scans, or not scan actually, point the web app that we create. in front of the wall. positioning wall, with the GPS coordinates. We can show all these things properly. A wall of fame Basically, What I'm trying to do here is In Riga Technical University A lot of things keep happening, like hackathons, events, conferences, and so on. So for this, I want to keep a track, not a track as in track. A wall of fame. So what do you think about it? And... Do you have any suggestions or something else, then you can tell me.

I think this is a strong idea for RTU. A wall of fame for events is more realistic than trying to force perfect indoor GPS tracking, and it fits the university's hackathons, conferences, and student projects very well.

What works best

For a huge hall, I would not try to make the experience depend on exact indoor positioning alone, because regular GPS does not work indoors and AR.js location-based AR is only low-precision indoors. Instead, the user should stand at a marked viewing zone, point the phone at the wall, and the app should use the wall's stored GPS anchor plus a calibration step to show the right content area. ^[48] ^[49]

That gives you the effect you want without requiring wall markers on the display itself. The wall becomes a digital showcase surface: events, winners, projects, photos, videos, and maybe a timeline of RTU innovation. ^[50] ^[51]

My recommendation

Make it a **hybrid wall of fame**, not just one AR screen. The hall should have:

- a main “featured event” panel.
- a scrolling timeline of hackathons and conferences.
- winner cards with team names, project titles, and year.
- optional QR/open-web links for details when users tap a card. ^[51] ^[50]

If the wall is in a fixed hall, you can pre-register one or more “anchor zones” for each section of the wall. Then the app checks the user's rough location and orientation, and when they are in the right spot it reveals the corresponding panel. That is much more stable than expecting GPS to precisely lock onto the wall indoors. ^[49] ^[48]

Better positioning idea

Instead of one exact coordinate, define a small **zone** in front of the wall:

- one coordinate for the center of the viewing area.
- a radius of a few meters.
- a heading range so the user is facing the wall.
- a “ready” state that only activates when both conditions are close enough. ^[48] ^[49]

That would feel like standing in front of the wall and seeing the content appear naturally, which is probably the best version of your concept for a hall demo. For indoor recognition and precise placement, real indoor positioning systems or visual alignment are usually better than pure GPS. ^[52] ^[48]

Extra suggestions

I would add these features:

- Event filters: hackathons, research, conferences, startups.
- Year filter: 2024, 2025, 2026.
- “Featured RTU moment” cards for major wins.

- A submission form so organizers can add new achievements later.
- A fallback web mode so the wall still works without AR on desktop. ^[53] ^[50]

This could become an RTU “digital legacy wall” as well as an event showcase, which makes it more valuable long term than only a hackathon-only display. ^[50] ^[51]

Practical caution

If you want this to feel truly reliable in the hall, you should treat GPS as a coarse trigger and not the final alignment mechanism. Indoor GPS is weak, Wi-Fi/BLE/UWB improve positioning, and AR-style indoor anchors can help, but the best user experience still comes from combining location, orientation, and a fixed viewing zone. ^[54] ^[52] ^[48]

Would you like me to turn this into a **real RTU wall-of-fame web prototype** with sections for hackathons, conferences, and winners, plus a “stand here” calibration screen?

✱✱

Yes, it is a fixed wall, like it is a huge hall where conferences are being held, where hackathons are being held. So, yes, we can have some anchor zones. For each action of the ball. Begin how GPS plus The local positioning of the wall. So I think we can use both, right? What do you think about it? We can have One zone for hackathons, one zone for conferences, one zone for events. And so on. And yes, you're right. I should treat GPS as a coarse trigger. And not the final alignment mechanism. Indoor GPS would be weak. But what we can do is Use latitude, longitude. of the phone. We can use GPS coordinates. And We can have AR-style indoor anchors. That can help In the fixed viewing zone. So tell me how we can do that? I mean, do I need to go there? Do I need to send you the pictures? Or just do one thing for now. in detail. Give me the plan. The detailed prompt All the technical things that we discussed. And a short idea. Description of this.

Yes — the right approach is to use GPS as a coarse trigger and then use an indoor wall anchor or local alignment step to make the content sit on the fixed hall wall. AR.js explicitly says location-based AR can work indoors but with low precision, and its own issue discussions note that GPS alone does not anchor content robustly to the environment. ^[64] ^[65]

How I would design it

Use a two-layer system:

1. Coarse zone detection from GPS:

- Check whether the user is inside the hall's approximate latitude/longitude zone.
- Confirm the user is facing the wall with device heading.
- If true, unlock the relevant wall section.

2. Local wall alignment inside the hall:

- Use a fixed viewing zone on the floor.
- Use device orientation plus a small manual calibration step.

- Optionally use a room map or AR world map if you later move beyond plain web AR. ^[66] ^[67] ^[64]

That gives you the “no marker on the wall” behavior you want, while still making the wall feel stable enough for a hall demo. GPS is then only used to decide *which wall zone* should appear, not to perfectly pin the content. ^[68] ^[64]

Zone idea

Your idea of multiple zones is solid:

- Hackathons zone.
- Conferences zone.
- Events zone.
- Alumni or research zone.

Each zone can have:

- a center coordinate,
- a radius,
- a valid heading range,
- and a content panel template. ^[69] ^[64]

For example, if someone stands in the right area and points at the wall, the app can show the “Hackathons” wall, while another spot in the hall can unlock “Conferences.” That makes the hall behave like a living digital wall of fame rather than a single static AR object. ^[70] ^[71]

What I need from you later

You do not need to send pictures yet to plan the system. For now, I can give you the architecture and a detailed build prompt; later, if you want exact alignment, photos of the hall wall and a rough floor plan will help refine the viewing zone and layout. Indoor AR systems become much better when the wall geometry and user standing position are known, but you can start with coordinate zones first. ^[67] ^[66]

Short concept description

RTU Wall of Fame is a fixed indoor web AR installation for the university hall that reveals event, hackathon, and conference highlights when a user stands in a predefined viewing zone and points the phone at the wall. It uses GPS only as a coarse location check and combines that with local wall alignment to create a stable digital showcase. ^[64] ^[70]

Detailed build prompt

Here is a prompt you can use to build the system:

Build a mobile-first WebAR “RTU Wall of Fame” prototype for a fixed university hall wall. The experience should work indoors without wall markers, using GPS as a coarse trigger and a local wall viewing zone for alignment. The hall contains multiple anchor zones: Hackathons, Conferences, Events, and Research. Each zone has its own latitude, longitude, radius, and heading range. When the user enters the zone and points the phone at the wall, the app reveals a large AR panel on the wall with event cards, winner names, project titles, year, and media links. Use AR.js / A-Frame style location-based AR as the baseline, but do not rely on GPS for exact wall placement. Add a calibration overlay instructing the user to stand at a fixed spot and face the wall. Add a status indicator showing “Searching for zone,” “Zone found,” and “Aligned.” Make the design elegant, clean, and suitable for a university conference hall. Include a fallback desktop web view that shows the same wall-of-fame content in a normal scrollable layout. Use sample RTU event data and make it easy to replace the data source later with JSON or Google Sheets. Optimize for fast load, phone browsers, and presentation/demo use.

Technical plan

- Define hall zones in a JSON array with coordinates, radius, and heading range.
- Use `navigator.geolocation` to get the user position.
- Use device orientation to estimate whether the user is facing the wall.
- Show a ready state only when both position and heading match the zone.
- Render the wall content as a large panel in the camera view.
- Keep the content modular so each zone loads a different set of event cards.
- Add an admin-friendly data file so RTU staff can update winners and event lists easily. ^[69]
_{[68] [64]}

Next step

The best next step is for me to turn this into a **full prototype spec + page structure + data model** for the wall of fame, and then I can generate the actual HTML/JS version around that.



1. https://www.reddit.com/r/augmentedreality/comments/1ai41p4/is_this_possible_ar_without_markers_and_without/
2. <https://www.wintor.com/features>
3. <https://idevbrandon.medium.com/location-based-ar-with-ar-js-d1ff05e55dc7>
4. <https://www.coursera.org/articles/augmented-reality-framework>
5. <https://developers.google.com/ar>
6. <https://ar-js-org.github.io/studio/pages/location/index.html>
7. <https://www.banuba.com/blog/what-is-augmented-reality>
8. <https://www.facebook.com/MAGES.Institute/posts/some-of-the-best-innovations-are-brought-about-by-bring-two-different-technologi/786901973463320/>
9. <https://d-nb.info/1215841752/34>

10. <https://www.facebook.com/groups/995657157195251/posts/2788577951236487/>
11. <https://www.algoryte.com/augmented-reality-services/>
12. <https://medium.com/@vivianlii/build-location-based-ar-on-the-web-using-ar-js-b9b9ea006819>
13. https://www.reddit.com/r/augmentedreality/comments/1bhljw2/geospatial_ar_in_metaspark_or_webbased/
14. <https://stackoverflow.com/questions/68023564/ar-js-location-based-not-placing-at-coordinates>
15. <https://www.scribd.com/presentation/887160454/Module-4-Introduction-to-Augmented-Reality>
16. https://www.reddit.com/r/augmentedreality/comments/1ai41p4/is_this_possible_ar_without_markers_and_without/
17. <https://hpc-portal.eu/node/1170>
18. <https://www.latlong.net/place/riga-latvia-13221.html>
19. <https://aframe.io/blog/arjs3/>
20. https://www.reddit.com/r/augmentedreality/comments/1ai41p4/is_this_possible_ar_without_markers_and_without/
21. <https://github.com/jeromeetienne/AR.js/issues/623>
22. <https://latitude.to/articles-by-country/lv/latvia/131269/riga-central-station>
23. <https://erasmusplay.com/en/riga/rigas-tehniska-universitate.html>
24. <https://github.com/AR-js-org/AR.js/issues/33>
25. https://en.wikipedia.org/wiki/Riga_Technical_University
26. <https://time-ok.com/coordinates/riga>
27. <https://www.topuniversities.com/universities/riga-technical-university>
28. <https://wiki.openstreetmap.org/wiki/Riga>
29. <https://latitude.to/articles-by-country/lv/latvia/38010/riga-technical-university>
30. <https://studyinlatvia.lv/universities/riga-technical-university>
31. <https://arcticstartup.com/event-location/riga-technical-university-kipsala-campus/>
32. <https://mapcarta.com/W215529684>
33. <https://ar-js-org.github.io/AR.js-Docs/location-based/>
34. <https://ieeexplore.ieee.org/iel8/6287639/10820123/10930457.pdf>
35. <https://stackoverflow.com/questions/72139386/location-based-ar-position-is-not-accurate>
36. <https://medium.com/@jcrueda/vr-positional-tracking-experiments-with-ar-js-a-frame-i-single-marker-scene-7e034224515c>
37. <https://www.sciencedirect.com/science/article/pii/S1226798824051511>
38. https://www.reddit.com/r/webdev/comments/1ehq3hk/need_tech_advice_for_building_an_indoor/
39. <https://www.zappar.com/insights/6-powerful-ways-to-leverage-location-based-ar>
40. <https://github.com/FireDragonGameStudio/ARIndoorNavigation-Threejs>
41. <https://developers.google.com/ar/develop/geospatial>
42. <https://www.multiset.ai/post/top-ar-indoor-navigation-tools-platforms-of-today>
43. <https://pmc.ncbi.nlm.nih.gov/articles/PMC8444080/>
44. <https://github.com/AR-js-org/AR.js/issues/33>
45. <https://dl.acm.org/doi/10.1145/3544549.3585745>

46. <https://www.indooratlas.com/blog/a-unique-blend-of-augmented-reality-ar-and-indoor-positioning/>
47. <https://github.com/google-ar/arcore-unity-extensions/issues/211>
48. https://marvelmind.com/indoor_gps/
49. <https://ar-js-org.github.io/AR.js-Docs/location-based/>
50. <https://www.nevco.com/modernize-your-schools-history-and-legacy-with-a-digital-wall-of-fame/>
51. <https://www.intuiface.com/blog/how-to-create-an-interactive-hall-of-fame-for-your-school>
52. <https://situm.com/en/>
53. <https://digitalrecordboard.com/blog/how-to-build-digital-wall-of-achievement/>
54. <https://tgtracking.co.za/wifi-positioning-for-indoor-tracking/>
55. <https://www.facebook.com/LSMnews/posts/the-first-riga-technical-university-rtu-hackathon-organized-in-cooperatio-n-with-/1132831248842425/>
56. <https://labsoflatvia.com/en/news/head-of-rtu-design-factory-receives-the-world-open-innovation-conference-prize>
57. <https://halloffame-online.com/digital-wall-of-fame/>
58. <http://www.wifarer.com/stadiums.html>
59. <https://vipinteract.com/digital-donor-wall/digital-hall-of-fame/>
60. <https://www.orient.me/8-unexpected-ways-to-use-indoor-gps/>
61. <https://halloffamewall.com>
62. <https://eng.lsm.lv/article/society/education/06.01.2025-student-hackathon-comes-up-with-idea-for-deadly-icicle-prevention.a582460/>
63. <https://estonianworld.com/technology/garage48-hold-hackathon-riga/>
64. <https://ar-js-org.github.io/AR.js-Docs/location-based/>
65. <https://github.com/AR-js-org/AR.js/issues/490>
66. <https://developer.apple.com/forums/thread/721764>
67. <https://www.indooratlas.com/blog/a-unique-blend-of-augmented-reality-ar-and-indoor-positioning/>
68. <https://stackoverflow.com/questions/72139386/location-based-ar-position-is-not-accurate>
69. <https://forum.playcanvas.com/t/location-based-ar-without-8th-wall/21573>
70. <https://www.nevco.com/modernize-your-schools-history-and-legacy-with-a-digital-wall-of-fame/>
71. <https://www.intuiface.com/blog/how-to-create-an-interactive-hall-of-fame-for-your-school>
72. <https://aframe.io/blog/arjs3/>
73. <https://forum.8thwall.com/t/indoor-3d-object-anchors/4512>
74. <https://github.com/jeromeetienne/AR.js/issues/623>
75. https://www.reddit.com/r/augmentedreality/comments/1k2fuh4/arjs_location_based_3d_model_keeps_following/
76. <https://www.youtube.com/watch?v=NbSqmZFROIU>
77. <https://www.rapid.com/en-gb/blog/rapid-tools/plasterboard-fixing/how-hollow-wall-anchors-work-and-how-to-use-them/>
78. <https://www.npmjs.com/package/ar.js?activeTab=readme>
79. <https://thediypaybook.com/wall-anchors-101/>
80. <https://stackoverflow.com/questions/76405749/issues-with-ar-js-location-based-example-from-documentation>

